

Key Business Insights & Analysis

1. Overall Shipment Performance

The logistics operation processed **25000 total shipments**, covering approximately **[37] Lakh KM** of delivery distance. The average shipment distance was **[3.67] KM**, with an average delivery cost of ₹ **[864.94] per shipment**.

Insight: This provides an overall view of shipment volume, operational scale, and the cost associated with fulfilling deliveries.

2. Delivery Timeliness & Delay Analysis

The overall **on-time delivery** was **[18000]**, while **[6669] of shipments were delayed**.

The analysis of delay patterns across different weather conditions showed that **[Stormy Weather]** recorded the highest delay rate at **[41.45] %**, while **[Cold Weather]** recorded the lowest at **[16.02] %**.

Insight: Delay performance varies across operating conditions, highlighting areas where logistics operations may require closer monitoring.

3. Regional Shipment Analysis

The **[West Region]** region handled the highest shipment volume with **[5095] shipments**, representing **[20.38]%** of total shipments.

The **[South Region]** region recorded the **[lowest ₹ 858.30] average delivery cost**, at approximately **[3.67] KM**.

Insight: Regional analysis helps identify shipment concentration and differences in operational requirements across geographical areas.

4. Vehicle Utilization Analysis

Among the available vehicle types, [**EV Bike**] handled the highest number of shipments with [**4218**] **deliveries**, while [**EV Van**] recorded the lowest volume with [**4116**] **deliveries**.

The lowest delay rate for [**Scooter**] was [**26.14**]%.

Insight: Vehicle-level analysis provides visibility into fleet utilization and differences in delivery performance.

6. Delivery Partner Performance

Delivery partners were evaluated using three key dimensions:

- Average Delivery Cost
- Average Delivery Rating
- Cost Efficiency

A weighted **Overall Performance Score** was developed to compare delivery partners on a standardized **0–100 scale**.

The highest-performing partners based on this custom index were [**Delhivery**], [**Fedex**], [**Dhl**], [**Ekart**], and [**Amazon Logistics**].

Insight: Combining multiple operational metrics into a single index provides a consistent framework for benchmarking delivery-partner performance.

7. Shipment Stress Analysis

A custom **Shipment Stress Index** was developed using:

- Delay Rate — **40%**
- Cost Pressure — **25%**

- Distance Pressure — **20%**
- Package Weight Pressure — **15%**

The highest shipment stress was observed in **[50 + KM / Express]**, with a stress index of **[61.81 %]**.

Insight: The index helps identify shipment segments where multiple operational pressures—delays, cost, distance, and package weight—are relatively elevated